

Declarations and author statement

CRediT author statement

Yuanyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft, Writing - review and editing, and Project administration.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data and code availability

The code and data supporting the findings are provided in the accompanying supplementary artifact submitted with the manuscript. The artifact includes source code, fixed test vectors, experiment configurations, raw JSON/JSONL results, generated tables and figures, TLA+ models, dependency versions, and quick/full reproduction scripts.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the author used AI-assisted coding and language tools to help develop executable reference models, organize experiments, inspect consistency, and edit prose. The author remains responsible for the research design, implementation, evidence, citations, interpretation, and final manuscript.

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None.